

Lab 05 — Path Traversal, Paranoia Levels & the Threshold Dial

Level: Intermediate **Goal:** Block path traversal, then learn the two dials an operator turns in production — **paranoia level** and **anomaly threshold** — and the trade-off they control. **Time:** ~40 minutes **Operator focus:** This is where you stop being a user of CRS and start *configuring* it. These two settings are the first thing you decide on any real deployment.

1. The attack — path traversal / LFI

Our `/file` endpoint reads a file from a `data/` directory:

```
target = os.path.join(base, name)  # name comes straight from the user
open(target)
```

If `name` is `report.txt`, fine. If `name` is `../../../../../../../../etc/passwd`, the `../` sequences climb out of `data/` and up to the filesystem root. Hit the **unprotected** app:

```
curl "http://localhost:5000/file?name=../../../../../../../../etc/passwd"
```

```
No WAF (:5000): path traversal reads /etc/passwd off the server [HTTP 200]
```

```
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
proxy:x:13:13:proxy:/bin:/usr/sbin/nologin
www-data:x:33:33:www-data:/var/www:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
list:x:38:38:Mailing List Manager:/var/list:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
_apt:x:42:65534:/nonexistent:/usr/sbin/nologin
nobody:x:65534:65534:nobody:/nonexistent:/usr/sbin/nologin
ubuntu:x:1000:1000:Ubuntu:/home/ubuntu:/bin/bash
systemd-network:x:998:998:systemd Network Management:/usr/sbin/nologin
messagebus:x:100:101:/nonexistent:/usr/sbin/nologin
redis:x:101:103:/var/lib/redis:/usr/sbin/nologin
postgres:x:102:104:PostgreSQL administrator,,,:/var/lib/postgresql:/bin/bash
polkitd:x:997:997:User for polkitd:/usr/sbin/nologin
claude:x:999:1001:/home/claude:/bin/bash
```

The server just handed a remote attacker its user database. This is **Local File Inclusion (LFI)** / **path traversal** — OWASP calls it Broken Access Control / Injection depending on the variant.

Through the WAF it's dead on arrival:

```
curl -i "http://localhost:8080/file?name=../../../../../../../../etc/passwd" # 403
```

Via WAF (:8080): path traversal BLOCKED (CRS 930xxx) [HTTP 403]

403 Forbidden

nginx

Rules `930100` / `930110` / `930120` (the LFI family) recognise the `../` and well-known target paths (`/etc/passwd`, `/proc/...`).

Real fix (dev side): never build a path from raw input — canonicalise and confirm it stays inside an allow-listed base directory (`os.path.realpath` + prefix check), or map user input to an ID, not a path. The WAF is the net for the endpoints you haven't hardened yet.

2. Dial #1 — Paranoia Level (how many rules run)

This is the single most important operator decision in CRS. The **paranoia level (PL)**, set in `crs-setup.conf` (or `PARANOIA` env var in this image), runs from **1 to 4** and controls **how aggressive** the rule set is.

PL	Character	What it adds	Who runs it
1	Default, minimal false positives	Well-proven rules only	Almost everyone, in production

PL	Character	What it adds	Who runs it
2	Stricter	More SQLi/XSS heuristics, stricter char checks	Mature deployments after tuning
3	Aggressive	Catches evasions, but noisy	High-security apps, well-tuned
4	Paranoid	Blocks lots of borderline input	Locked-down internal apps; expect heavy tuning

The higher the PL, the **more rules evaluate each request** — so you catch more evasions, but you also flag more legitimate traffic.

Why this matters: A higher paranoia level is *not* "just better security." Each level up trades false negatives for **false positives**. Turning it up without doing the tuning work (Lab 07) means blocking real customers. In industry the standard path is: **deploy at PL1, get clean, then consider PL2.**

Try it

Start WAFs at PL1, PL2 and PL4:

```
for pl in 1 2 4; do
  docker run -d --name modseclabs-waf-pl$pl \
    --add-host=host.docker.internal:host-gateway \
    -e BACKEND="http://host.docker.internal:5000" -e PORT=8080 \
    -e MODSEC_RULE_ENGINE=On -e PARANOIA=$pl -e ANOMALY_INBOUND=5 \
    -e MODSEC_AUDIT_LOG=/dev/stdout -e MODSEC_AUDIT_ENGINE=RelevantOnly \
    -p 808$pl:8080 owasp/modsecurity-crs:nginx
done
```

Now watch a **legitimate** product search get more suspicious as PL rises. Some benign strings are caught even at PL1 — a classic false positive:

```
curl -s -o /dev/null -w "%{http_code}\n" --get "http://localhost:8081/search" \
  --data-urlencode "q=Compare and select from our items"
```

That returns **403** — the phrase "select ... from" tripped the SQLi rules even though it's an honest search. **This is the false positive you will spend real time fixing in Lab 07.** It's not a bug; it's the fundamental tension of pattern matching, and managing it is the operator's core job.

3. Dial #2 — Anomaly Threshold (how much suspicion = block)

Recall from Lab 02: rules add points; rule `949110` blocks when the total crosses the **inbound anomaly threshold** (`SecAction ... setvar:tx.inbound_anomaly_score_threshold`, or `ANOMALY_INBOUND` here). Default = **5**, which equals **one CRITICAL rule**.

Threshold	Effect
5 (default)	One critical hit blocks. Strict.
10	Needs ~two critical hits. More forgiving; fewer false positives.
20+	Very permissive; only egregious multi-rule attacks blocked.

Operators often **pair** the two dials: raise the paranoia level for coverage, but also raise the threshold so a single borderline rule doesn't block on its own. For example "PL2 with threshold 10" is a common, well-balanced production posture.

Try it

Run one WAF at PL1 with a *relaxed* threshold of 15:

```
docker run -d --name modseclabs-waf-thresh \
  --add-host=host.docker.internal:host-gateway \
  -e BACKEND="http://host.docker.internal:5000" -e PORT=8080 \
  -e MODSEC_RULE_ENGINE=On -e PARANOIA=1 -e ANOMALY_INBOUND=15 \
  -e MODSEC_AUDIT_LOG=/dev/stdout -e MODSEC_AUDIT_ENGINE=RelevantOnly \
  -p 8090:8080 owasp/modsecurity-crs:nginx
sleep 6

# A payload that scores ~5 (one rule) now PASSES because threshold is 15:
curl -s -o /dev/null -w "relaxed(15) -> %{http_code}\n" \
  --get "http://localhost:8090/search" --data-urlencode "q=Compare and select from our items"
```

Same request that was **403** at threshold 5 may now be **200** at threshold 15 — you traded strictness for fewer false positives, globally, with one number.

The operator's mental model: *Paranoia level* decides **which** rules ask questions; *threshold* decides **how many** "yes, suspicious" answers it takes to block. Tuning a WAF is mostly moving these two dials and then surgically excluding the specific rules that still misfire (Lab 07).

4. Where these live in a real config

In a non-Docker install (or when you mount your own config), these are plain directives you version-control:

```
# crs-setup.conf
SecAction "id:900000, phase:1, nolog, pass, t:none, \
  setvar:tx.blocking_paranoia_level=1"

SecAction "id:900110, phase:1, nolog, pass, t:none, \
  setvar:tx.inbound_anomaly_score_threshold=5, \
  setvar:tx.outbound_anomaly_score_threshold=4"
```

The Docker image simply generates these from the `PARANOIA` and `ANOMALY_INBOUND` env vars at startup. **Knowing the underlying directives matters** because in production you'll often mount a hand-edited `crs-setup.conf` rather than rely on env vars.

Clean up the extra containers

```
docker rm -f modseclabs-waf-pl1 modseclabs-waf-pl2 modseclabs-waf-pl4 \
modseclabs-waf-thresh 2>/dev/null
```

5. Recap

- Path traversal (`930xxx`) escapes a directory with `../`; the WAF blocks the pattern before your file-reading code runs.
- **Paranoia level (1–4)** = how many rules run = coverage vs false positives. **Start at PL1 in production.**
- **Anomaly threshold** = how much accumulated suspicion triggers a block. Raise it to be more forgiving.
- These two directives live in `crs-setup.conf` and are the operator's primary dials. Turning paranoia up **requires** tuning work — which is Lab 07.

Next: Lab 06 — you stop *configuring* CRS and start *writing your own rules* in the ModSecurity rule language (`SecRule`).